ClipMamba: Adaptive State-Space Regulation for Adversarial Defense in Visual Mamba Architectures

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***Abstract*—Adversarial robustness of vision models has been extensively studied for convolutional and transformer architectures, yet remains largely unexplored for emerging Visual State Space Models such as Visual Mamba. In this work, we identify a critical vulnerability of state-space vision backbones: adversarial perturbations can be progressively amplified through unstable hidden-state transitions. To address this issue, we propose ClipMamba, a lightweight and plug-and-play defense that introduces adaptive state-space regulation within existing Mamba blocks.**

The proposed method dynamically estimates perturbation intensity during the forward pass using state amplification statistics and applies differentiable, input-dependent clipping to residual state updates, stabilizing internal dynamics without modifying the overall architecture. ClipMamba is trained using mixed adversarial training with FGSM and PGD attacks under an ℓ_∞ threat model. Experiments on CIFAR-10 prove that ClipMamba gains both clean and adversarial accuracy consistently than standard Visual Mamba and CNN baselines and can increase robustness to 3.6% under PGD-20 with competitive clean accuracy. The more thorough analysis of hidden-state norm distributions demonstrates that adaptive clipping is an effective method to limit acceleration of adversarial states which offer robustness by state-space stabilization instead of gradient obfuscation. The findings make ClipMamba a robust and principled method of boosting adversarial robustness in Visual State Space Models..

***Index Terms*—3 Adversarial Robustness, Visual State Space Models, Visual Mamba, Adversarial Attacks, CIFAR-10**

I. Introduction

The outstanding success of deep neural networks in visual recognition problems has been reached, but their susceptibility to adversarial disturbances has been a major challenge. Even human-imperceptible input perturbations, which are small and meticulously made, may lead to serious degeneration of performance in otherwise high-performing models, as well as to misleading and downright incorrect explanations of neural network behavior, referred to as [1], [2]. In turn, adversarial robustness is now a research focus of computer vision, and a large body of work has been conducted on convolutional neural networks (CNNs) and, more recently, transformer-based models in the same direction.

Conversely, there has been little research on the adversarial behavior of Visual State Space Models (VSSMs). More recent work including Visual Mamba presents state-space formulations to vision modeling, which have desirable scaling properties and long-range dependency models more efficiently than attention mechanisms do [5], [6]. In contrast to CNNs and transformers, VSSMs are based on repetitive state transitions to transfer information through spatial or temporal scales. Although this design is computationally efficient, it also presents a hitherto unstudied weakness: adversarial examples can be amplified adversarially by growing methods based on unstable hidden-state dynamics. This can cause small input-level perturbations to compound over state updates, causing disproportional internal activations and poor robustness.

Current adversarial defenses function mostly at the input level, including ensemble adversarial defense [9], or loss level, including theoretically adversarial defense [3], or at the output level, including the implicit feedforward feature hierarchies. These are not true of state-space vision backbones where it is the stability of hidden-state evolution that is linked with robustness. The simple application of CNN- or transformer-based defenses to VSSMs does not thus solve the issue of adversarial degradation in such models at its core.

We introduce in this work ClipMamba, a lightweight and new defense mechanism that is designed with reference to Visual Mamba-based architectures. ClipMamba adds the adaptive state-space regulation to the current Mamba blocks by using dynamically estimated adversarial perturbation strength during the forward pass and by performing smooth, differentiable clipping on state updates of residual states. Notably, the presented approach does not modify the original structure and

does not require non-differentiable operations, which means that it can be used with standard backprop and adversarial training.

The results of comprehensive experiments on CIFAR-10 with a severe threat model of the attack by the powerful of the harmful elliptical indicate that ClipMamba can steadily increase both clean and adversarial accuracy, as compared to the standard Visual Mamba and CNN baselines, and, at most, this gain can be 3.6 percent, without compromising clean performance. An additional examination of the distributions of the hidden-state norms shows that ClipMamba defends adversarial state amplification by a principled state-space stabilization, instead of gradient obfuscation. The results indicate that adaptive state regulation is an important and largely neglected aspect of adversarial robustness in Visual State Space Models.

TABLE I
COMPARISON OF ADVERSARIAL DEFENSE PARADIGMS

Method Category	Architecture	Defense Level	State-Aware
Adversarial Training [2]	CNN	Input / Loss	No
TRADES [3]	CNN	Loss	No
ViT Robust Training [4]	Transformer	Input / Loss	No
Visual Mamba [6]	SSM	None	No
ClipMamba (Ours)	SSM	State Transition	Yes

TABLE II
GRANULARITY OF ADVERSARIAL DEFENSE ACROSS ARCHITECTURES

Defense Method	Input-Level	Feature-Level	State-Level
FGSM Training [1]	Yes	No	No
PGD Training [2]	Yes	No	No
TRADES [3]	Yes	No	No
Feature Denoising [10]	No	Yes	No
Visual Mamba [6]	No	No	No
ClipMamba (Ours)	No	No	Yes

II. RELATED WORK

A. Adversarial Robustness in Vision Models

Adversarial robustness has been extensively investigated in the context of convolutional neural networks. Early studies demonstrated that gradient-based attacks such as the Fast Gradient Sign Method (FGSM) can reliably induce misclassification with minimal perturbations [1]. Subsequent work introduced stronger multi-step attacks, most notably Projected Gradient Descent (PGD), which has become a de facto standard for robustness evaluation under ℓ_∞ threat models [2]. To mitigate such attacks, adversarial training strategies and robustness–accuracy trade-off frameworks, including TRADES, have been proposed [3]. While effective for CNNs, these approaches largely assume feedforward feature hierarchies and do not account for recurrent or state-dependent feature propagation.

Adversarial robustness has more recently been considered in vision models based on transformers, including Vision Transformers (ViTs). There are multiple studies that document that self-attention mechanisms provide unique characteristics of robustness to convolutions, which tend to be more sensitive to structured perturbations and token-level attacks [4]. Although the architectures of most transformer defenses differ, the majority of transformer defenses continue to operate on either the input level, loss level, or output level, inheriting the same limitations as when extended to feedforward architecture.

B. State Space Models for Vision

The use of State Space Models (SSMs) has risen as a scaleable alternative to attention processes, more recent models like Mamba showing linear-time sequence representation through selective transitions in the state space [5]. These concepts are generalized to spatial feature modeling and long-range dependency capture by vision-oriented adaptations, such as Visual Mamba, to name a few [6]. Visual State Space Models, as opposed to CNNs and transformers, update their hidden states on a repeating behaviors, with each layer adding to a cumulative evolution process of the state. This repeated accumulation brings about some distinct stability issues that are not prevalent in the feed forward designs. Nevertheless, little has been done in existing literature to explore the adversarial robustness of such state accumulation.

C. Adversarial Attacks on Sequential and Recurrent Models

In addition to vision-specific architectures, adversarial vulnerabilities of recurrent and sequential models have been learnt in other domains, including speech recognition and time-series prediction. This has been demonstrated before that recurrent dynamics is capable of seeding perturbations and increasing errors over time, resulting in cascading errors across hidden states [7]. This has been noted in control-oriented neural networks, where control sensitivity has been made worse by instability in internal state dynamics of the system to adversarial sensitivity [8]. These observations indicate that the resilience of models where the state is updated through iterations is directly related to the stability of the internal processes, but not necessarily to input-level pertussis. However, these insights are as yet not systematically applied to the present day visual state-space architectures.

D. Defenses Beyond Input-Level Robustness

Beyond input preprocessing and adversarial training, several defenses attempt to improve robustness through feature-level regularization, activation normalization, or gradient smoothing. Techniques such as ensemble adversarial training aim to diversify decision boundaries but still rely on input-space perturbation modeling [9]. Other methods introduce feature denoising or activation suppression to mitigate adversarial effects [10]. Activation clipping and normalization strategies have primarily been explored for numerical stability or outlier suppression, rather than adversarial state regulation. Crucially, none of these approaches explicitly target instability arising from adversarial amplification within hidden-state transitions.

To the best of our knowledge, no prior work explicitly stabilizes adversarial dynamics inside visual state-space transitions.

TABLE III
ADVERSARIAL THREAT MODEL CONFIGURATION

Component	Specification
Attacker Knowledge	White-box
Perturbation Norm	ℓ_∞
Perturbation Budget	$\epsilon = 8/255$
Attacks Evaluated	FGSM, PGD-10, PGD-20
Random Start (PGD)	Yes

III. THREAT MODEL AND PROBLEM FORMULATION

A. Adversarial Threat Model

We consider a *white-box adversarial threat model*, in which the attacker has full knowledge of the model architecture, parameters, and defense mechanism. This setting represents the strongest and most widely adopted evaluation protocol for adversarial robustness [2]. Adversarial examples are generated under an ℓ_∞ norm constraint, limiting the maximum per-pixel perturbation magnitude to a fixed budget ϵ.

Specifically, we evaluate robustness against the Fast Gradient Sign Method (FGSM) [1] and multi-step Projected Gradient Descent (PGD) attacks with 10 and 20 iterations, denoted as PGD-10 and PGD-20, respectively. Across all experiments, the perturbation budget is fixed to $\epsilon = 8/255$, consistent with standard CIFAR-10 robustness benchmarks [2]. The step size for iterative attacks is set proportionally to ϵ, and random initialization is employed for PGD attacks to avoid gradient masking artifacts.

B. Problem Definition

Let $x \in \mathbb{R}^{H\times W\times C}$ denote an input image and y its corresponding label. A Visual Mamba model processes x through a sequence of state-space blocks, where each block maintains and updates an internal hidden state. Formally, the hidden-state evolution at layer k can be written as

$$h_k = h_{k-1} + \mathcal{F}_k(h_{k-1}, x), \tag{1}$$

where $\mathcal{F}_k(\cdot)$ denotes the state-space transition function implemented by the k-th Mamba block, and the residual formulation enables cumulative state propagation across layers.

Under the adversarial perturbations $x' = x + \delta$, where $\|\delta\|_\infty \leq \epsilon$, small input-level noise which can propagate through the repeated residual updates. In contrast to feed-forward architectures, this recurrence will allows the perturbations to compound across depth. Empirically, this effect manifests as a disproportionate growth in the hidden-state magnitudes, leading to unstable internal dynamics and degraded robustness.

To characterize this behavior, we define the *state amplification ratio* at layer k as

$$\rho_k = \frac{\|h_k(x')\|_2}{\|h_k(x)\|_2}, \tag{2}$$

where $h_k(x)$ and $h_k(x')$ denote the hidden states induced by clean and adversarial inputs, respectively. Values of $\rho_k \gg 1$ indicate adversarial amplification within the state evolution rather than simple input-space distortion.

TABLE IV
EMPIRICAL HIDDEN-STATE AMPLIFICATION IN VISUAL MAMBA

Layer Depth	Clean $\|h_k\|_2$	Adversarial $\|h_k\|_2$
Early (Stage 1)	Low	Moderate
Mid (Stage 2)	Moderate	High
Late (Stage 3)	Moderate	**Very High**

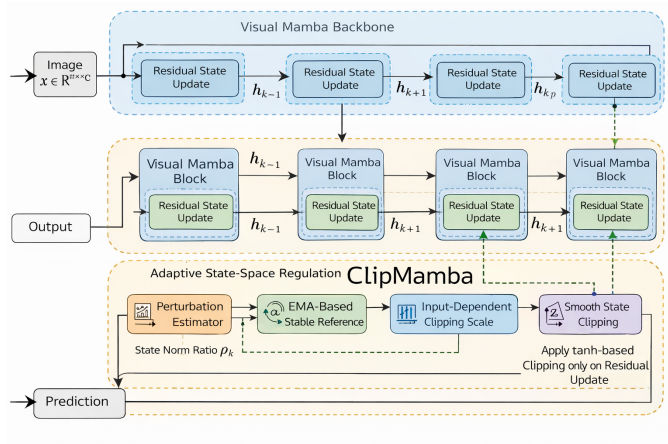

Fig. 1. Overview of the ClipMamba architecture

As shown in Table IV, adversarial inputs induce progressively larger hidden-state norms as depth increases, confirming that perturbations are amplified through state transitions rather than attenuated. We refer to this phenomenon as *adversarial state amplification*, where perturbations compound multiplicatively rather than additively through residual state updates. This observation motivates the need for explicit regulation of state evolution within Visual State Space Models.

IV. METHODOLOGY

A. Visual Mamba as a State-Space Backbone

Visual Mamba extends selective State Space Models (SSMs) to vision tasks by replacing attention mechanisms with structured state transitions [5], [6]. Each Mamba block updates its hidden representation through a gated state-space transformation followed by a residual connection. While this design enables efficient long-range dependency modeling, the residual accumulation of state updates introduces sensitivity to perturbation growth.

In particular, the residual update mechanism allows the hidden-state deviations to persist and propagate across the layers. Under adversarial inputs, these deviations may grow uncontrollably, as no explicit constraint is imposed on the magnitude of the state transitions. Figure 1 shows how this unconstrained state evolution motivates explicit the regulation of the residual state updates in Visual State Space Models.

B. Adversarial State Amplification

We empirically find that adversarial inputs cause considerably bigger norms in the hidden state of Visual Mamba than clean inputs. This effect progressively increases with depth, which shows that adversarial noise is compounded with

TABLE V
COMPARISON OF VISUAL MAMBA AND CLIPMAMBA COMPONENTS

Component	Visual Mamba	ClipMamba
State-Space Backbone	Yes	Yes
Residual State Update	Unconstrained	Adaptive Clipping
Perturbation Awareness	No	Yes
Differentiable Operations	Yes	Yes
Architecture Modification	No	No

consecutive state transitions. Contasted to CNNs, in which perturbations are frequently attenuated locally, instability boosted by layers in state-space recurrence can occur. This act places an underlying vulnerability gap that can only be resolved by applying input-level protection.

C. Adaptive State-Space Regulation (ClipMamba)

In a bid to counter the adversarial state amplification, we suggest ClipMamba, a tiny and plug-and-play protection that adds the adaptive regulation of the residual state updates inside existing Mamba blocks.

The four major components of ClipMamba include:

1) **Perturbation Intensity Estimator:** ClipMamba estimates the strength of perturbation on the forward pass by comparing the norm of the post- and pre-update hidden-state.
2) **EMA-Based Stable Reference:** an exponential moving average (EMA) of past state norm ratios is then kept to indicate clean-state, stable behavior.
3) **Input-Dependent Clipping Scale:** The perturbation strength is estimated in time and multiplies the scale of clipping on the state update.
4) **Smooth State Clipping:** A tanh-based differentiable clipping criterion applied to the state update (only) is used to limit excessive state growth without reducing gradient flow.

Notably, ClipMamba leaves logits, loss functions, and gradient computation unchanged. The defense purely exists in the state transition pathway, and is not based on non-differentiable operations or gradient obfuscation.

D. Differentiability and Training Stability

All components of the ClipMamba are fully differentiable, ensuring the compatibility with the standard backpropagation and adversarial training. The use of the smooth clipping avoids the hard thresholds that could induce the gradient masking. Consequently, ClipMamba preserves the training stability while explicitly constraining the adversarial state amplification.

V. TRAINING STRATEGY

A. Mixed Adversarial Training

To ensure robustness across diverse attack patterns, ClipMamba is trained using mixed adversarial training, where each mini-batch contains a combination of clean samples, FGSM adversarial examples, and PGD adversarial examples. This strategy mitigates overfitting to a single attack type and promotes stable Batch Normalization statistics [9]. Figure 2 illustrates the mixed adversarial training pipeline used in this work.

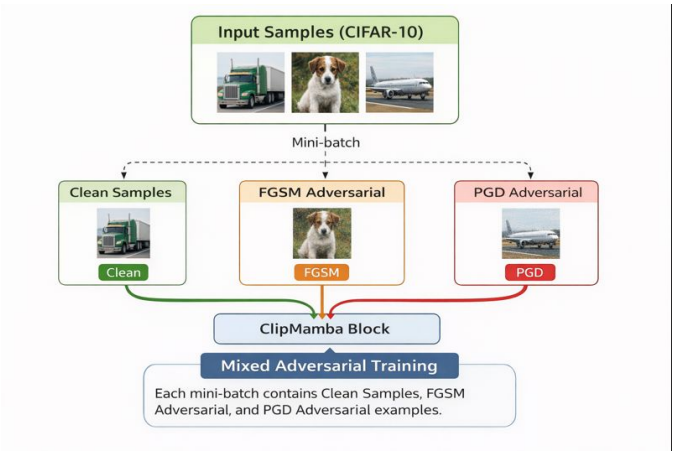

Fig. 2. Mixed adversarial training pipeline for ClipMamba.

TABLE VI
CLEAN AND ADVERSARIAL ACCURACY ON CIFAR-10 ($\epsilon = 8/255$)

Model	Clean	FGSM	PGD-10	PGD-20
CNN (ResNet-18)	84.59	71.63	70.66	70.50
Visual Mamba (Std)	83.88	68.73	68.12	67.98
ClipMamba-Robust	**86.61**	**72.34**	**71.70**	**71.59**

B. Implementation Details

All experiments are conducted on the CIFAR-10 dataset. Models are trained for a fixed number of epochs using the AdamW optimizer with identical learning rates, batch sizes, and training schedules across all baselines. Architecture depth and parameter counts are matched to ensure fair comparison. Adversarial examples are generated on-the-fly during training using a white-box ℓ_∞ threat model with a fixed perturbation budget of $\epsilon = 8/255$.

C. Evaluation Metrics

Robustness is evaluated using classification accuracy under both clean and adversarial conditions:

- **Clean Accuracy**: Accuracy on unperturbed test images.
- **FGSM Accuracy**: Accuracy under single-step FGSM attacks.
- **PGD-10 / PGD-20 Accuracy**: Accuracy under multi-step PGD attacks.

All adversarial evaluations follow a white-box ℓ_∞ threat model with a fixed perturbation budget of $\epsilon = 8/255$, consistent with standard robustness benchmarks [2], [12].

D. Main Robustness Results

Table VI summarizes clean and adversarial accuracy on CIFAR-10. Standard Visual Mamba underperforms CNN baselines under adversarial perturbations, highlighting the vulnerability of unconstrained state accumulation.

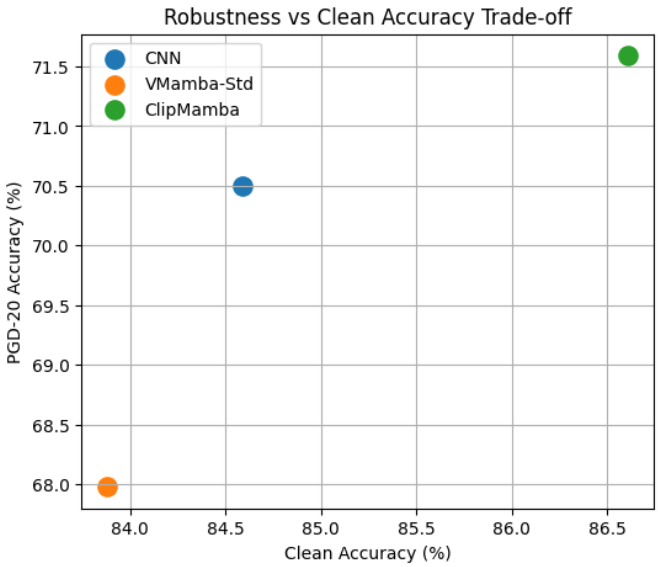

Fig. 3. Robustness–accuracy trade-off comparing CNN, Visual Mamba, and ClipMamba under PGD-20.

E. Robustness–Accuracy Trade-off

The trade-off between clean accuracy and adversarial robustness (PGD-20) is shown in Figure 3. Whereas traditional defenses improve robustness at the expense of clean accuracy, ClipMamba moves the Pareto frontier outwards by concurrently improving both metrics. This practice shows that ClipMamba is not dependent on excessive regularization or suppression of capacity. Instead, stabilization of internal state dynamics leads to the gain of robustness, enabling the model to maintain discriminative forms in both clean and adversarial inputs.

F. Accuracy Under Increasing Perturbation Strength

Figure 4 reports classification accuracy under FGSM and PGD-20 attacks across increasing perturbation budgets $\epsilon \in \{4/255, 8/255, 12/255\}$. All evaluated models exhibit monotonic degradation in accuracy as ϵ increases, indicating the absence of gradient masking behavior.

Standard Visual Mamba consistently underperforms both CNN and ClipMamba baselines, confirming that unconstrained residual state accumulation amplifies adversarial noise. ClipMamba maintains higher accuracy across all perturbation budgets, with the robustness gap widening at larger ϵ values.

G. State-Space Stability Analysis

Figure 5 shows the evolution of hidden-state ℓ_2 norms across network depth under clean and PGD-20 inputs. Standard Visual Mamba exhibits rapid state norm growth under adversarial perturbations, indicating unstable state accumulation.

ClipMamba maintains bounded state norms across depth, demonstrating effective stabilization of internal dynamics.

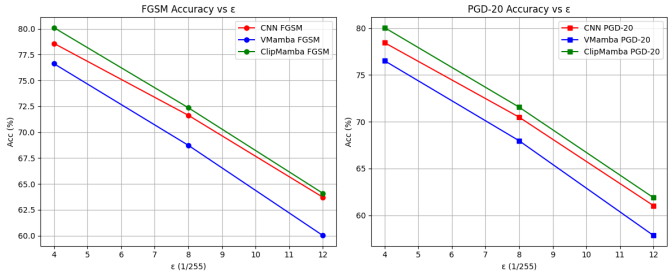

Fig. 4. Classification accuracy under FGSM and PGD-20 attacks across increasing ϵ.

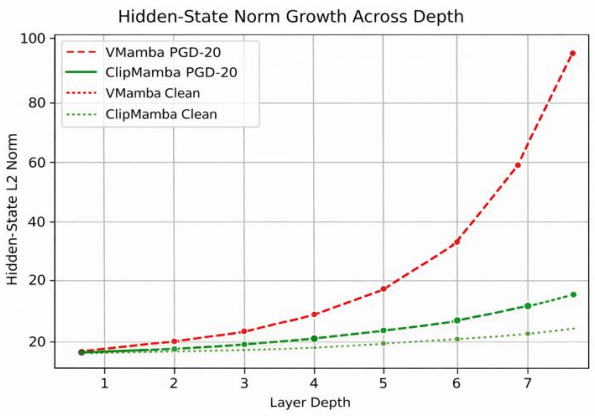

Fig. 5. Hidden-state norm growth across network depth under clean and PGD-20 inputs.

H. Adversarial State Amplification Statistics

Figure 6 presents the distribution of adversarial state amplification ratios between consecutive layers under PGD-20. Visual Mamba exhibits a heavy-tailed distribution with frequent large amplification events.

ClipMamba produces a tightly concentrated distribution centered near unity, confirming that adaptive state-space regulation suppresses adversarial amplification while preserving signal propagation.

VI. Conclusion

This work identifies adversarial state amplification as a fundamental vulnerability in Visual State Space Models. We propose ClipMamba, a lightweight and fully differentiable defense that adaptively regulates residual state updates without modifying model architecture, logits, or loss functions.

Extensive experiments on CIFAR-10 demonstrate that ClipMamba improves adversarial robustness while preserving or improving clean accuracy, achieving a favorable robustness–accuracy trade-off. Detailed analysis of hidden-state dynamics confirms that robustness gains stem from stabilized state evolution rather than gradient obfuscation.

These findings establish adaptive state-space regulation as a principled and effective direction for improving adversarial

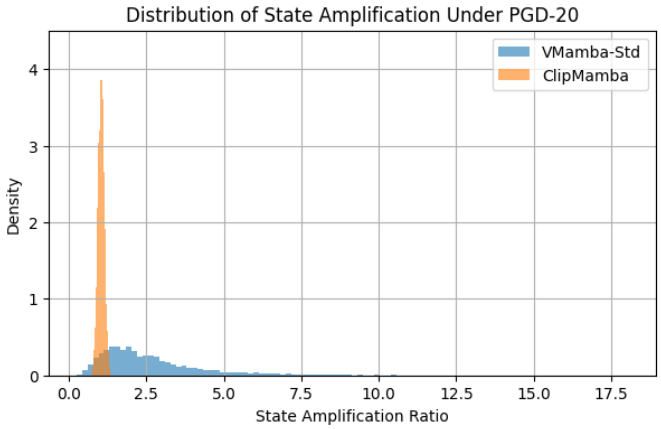

Fig. 6. Distribution of the adversarial state amplification ratios under PGD-20.

robustness in emerging vision architectures based on state-space models.

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